Class 10 Science
Chemical Reactions and Equations

PART 1 - NCERT EXERCISE SOLUTIONS (Q1-Q5)

Q1. Which statements are incorrect?

$2PbO + C \rightarrow 2Pb + CO_2$

(a) Lead oxide is reduced.

(b) Carbon is oxidised.

(c) Lead is oxidised.

(d) Carbon is reduced.

Answer:

- Lead oxide loses oxygen, so it is REDUCED.
- Carbon gains oxygen, so it is OXIDISED.

Correct option: (i) (a) and (b)

Q2. Identify the type of reaction

$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$

Answer:

This is a Displacement Reaction.

Correct option: (d)

Q3. What happens when dilute HCl is added to iron filings?

$Fe + 2HCl \rightarrow FeCl_2 + H_2\uparrow$

Answer:

- Hydrogen gas is produced.
- Iron chloride is formed.

Correct option: (a)

Q4. What is a balanced chemical equation? Why should it be balanced?

Need for balancing:

→ It follows the LAW OF CONSERVATION OF MASS.

→ Mass can neither be created nor destroyed.

Answer:

A chemical equation in which the number of atoms of each element is equal on both sides is called a BALANCED CHEMICAL EQUATION.

Q5. Translate into chemical equations and balance them.

(a) Hydrogen + Nitrogen → Ammonia → $N_2 + 3H_2 \rightarrow 2NH_3$

(b) Hydrogen sulphide burns in air → $2H_2S + 3O_2 \rightarrow 2H_2O + 2SO_2$

(c) Barium chloride + Aluminium sulphate → $3BaCl_2 + Al_2(SO_4)_3 \rightarrow 3BaSO_4 + 2AlCl_3$

(d) Potassium + Water → $2K + 2H_2O \rightarrow 2KOH + H_2$

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PART 2 - NCERT EXERCISE SOLUTIONS (Q6-Q10)

Q6. Balance the following chemical equations.

(a) $2HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + 2H_2O$

(b) $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

(c) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

(d) $BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2HCl$

Q7. Write balanced chemical equations for the following reactions.

(a) $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

(b) $Zn + 2AgNO_3 \rightarrow Zn(NO_3)_2 + 2Ag$

(c) $2Al + 3CuCl_2 \rightarrow 2AlCl_3 + 3Cu$

(d) $BaCl_2 + K_2SO_4 \rightarrow BaSO_4 + 2KCl$

Q8. Write the chemical equations for the following reactions and identify the type of reaction.

(a) $2KBr + BaI_2 \rightarrow 2KI + BaBr_2$

Type : Double displacement reaction

(b) $ZnCO_3 \rightarrow ZnO + CO_2$

Type : Decomposition reaction

(c) $H_2 + Cl_2 \rightarrow 2HCl$

Type : Combination reaction

(d) $Mg + 2HCl \rightarrow MgCl_2 + H_2$

Type : Displacement reaction

Q9. What are exothermic and endothermic reactions? Give one example of each.

Exothermic Reaction :

A reaction in which heat is released to the surroundings.

Example :

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + Heat$

(Combustion of methane)

Endothermic Reaction :

A reaction in which heat is absorbed from the surroundings.

Example :

$CaCO_3 \xrightarrow{Heat} CaO + CO_2$

(Thermal decomposition of calcium carbonate)

Q10. Why is respiration considered an exothermic reaction? Write the equation.

Answer: Respiration is an exothermic reaction because energy is released when glucose is oxidised in the presence of oxygen in our body.

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + Energy$

(Glucose) (Energy released)

PART 3 - NCERT EXERCISE SOLUTIONS (Q11-Q20)

Q11. Why are decomposition reactions opposite to combination reactions?

Answer:

- In combination reactions, two or more substances combine to form a single product.
- In decomposition reactions, a single compound breaks down into two or more simpler substances.

Examples :

Combination Reaction	Decomposition Reaction
$CaO + H_2O \rightarrow Ca(OH)_2$	$CaCO_3 \rightarrow CaO + CO_2$
Two reactants → One product	One reactant → Two products

Q12. Write decomposition reactions brought about by heat, light and electricity.

Answer:

- By Heat (Thermal decomposition):

$$CaCO_3 \xrightarrow{\text{Heat}} CaO + CO_2$$

- By Light (Photochemical decomposition):

$$2AgCl \xrightarrow{\text{Sunlight}} 2Ag + Cl_2$$

- By Electricity (Electrolytic decomposition):

$$2H_2O \xrightarrow{\text{Electricity}} 2H_2 + O_2$$

Q13. What is the difference between displacement reaction and double displacement reaction? Write one example of each.

Answer:

- Displacement Reaction :
  A more reactive element displaces a less reactive element from its compound.
  Example :
  $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
- Double Displacement Reaction :
  The positive ions and negative ions of two compounds exchange their partners.
  Example :
  $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

Q14. What happens when silver nitrate solution is added to copper metal?

Answer:

Copper displaces silver from silver nitrate solution.

$$Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$$

- A greyish-white deposit of silver is formed on copper.
- The blue colour of the solution changes to light blue.

Q15. Define precipitation reaction. Write one example.

Answer:

A reaction in which an insoluble solid (precipitate) is formed is called a precipitation reaction.

Example :

$$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 \downarrow + 2NaCl$$

(White precipitate of $BaSO_4$ is formed)

Q16. What are oxidation and reduction? Write one example of each.

Answer:

- Oxidation : Addition of oxygen or removal of hydrogen.
  Example : $2Cu + O_2 \rightarrow 2CuO$
- Reduction : Removal of oxygen or addition of hydrogen.
  Example : $CuO + H_2 \rightarrow Cu + H_2O$

Q17. A shiny brown element on heating in air becomes black. Identify the element and black compound.

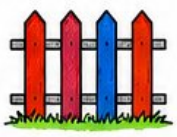
Answer:

- Element : Copper (Cu)
- Black compound : Copper(II) oxide (CuO)

Q18. Why do we paint iron articles?

Answer:

Painting prevents iron from coming in contact with air and moisture, thus preventing rusting (corrosion).



Q19. Why are food packets flushed with nitrogen?

Answer:

Nitrogen prevents oxidation of oils and fats present in the food. It prevents spoilage (rancidity) and increases the shelf life of food.



Q20. Define corrosion and rancidity.

Answer:



(a) Corrosion : Slow destruction of metals due to chemical reaction with air, moisture or other substances.
Example : Rusting of iron.

(b) Rancidity : Oxidation of fats and oils in food causing unpleasant smell and taste.
Example : Spoilage of potato chips, stale oil.



PART 4 – MOST IMPORTANT PREVIOUS YEAR QUESTIONS (PYQs)

1. What is corrosion? How can it be prevented?

Answer:

Corrosion is the slow damage or destruction of metals due to chemical reaction with air, moisture or other substances.

Prevention:



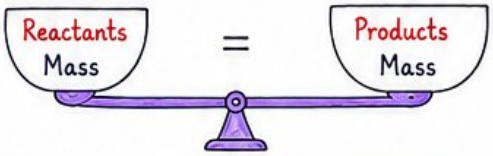
- Painting
- Greasing
- Galvanisation (coating with zinc)



2. Why should chemical equations be balanced?

Answer:

Chemical equations should be balanced to satisfy the Law of Conservation of Mass. It states that mass can neither be created nor destroyed in a chemical reaction.



3. Name the type of reaction.

$$2AgCl \xrightarrow{\text{Sunlight}} 2Ag + Cl_2$$

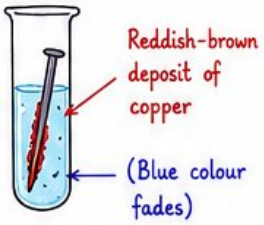
Answer:

This is a photochemical decomposition reaction.

4. What happens when an iron nail is dipped in copper sulphate solution?

Answer:

- The blue colour of copper sulphate solution fades (turns light green).
- A reddish-brown deposit of copper is formed on the iron nail.
- Iron gets coated with copper.



5. Why is respiration called exothermic?

Answer:

During respiration, glucose is oxidised in the presence of oxygen and energy is released. Since heat/energy is released, respiration is an exothermic reaction.

6. What is rancidity?

Answer:

Rancidity is the spoilage of food due to oxidation of fats and oils. It causes unpleasant smell and bad taste in food.



7. State one observation that helps us to identify a chemical reaction.

Answer:

Any one of the following observations helps us to identify a chemical reaction:

- Change in colour
- Evolution of a gas
- Change in temperature
- Formation of a precipitate



8. Define combination reaction with example.

Answer:

A combination reaction is a reaction in which two or more substances combine to form a single product.

Example:

$$CaO + H_2O \longrightarrow Ca(OH)_2$$

Two reactants — One product



9. Define decomposition reaction with example.

Answer:

A decomposition reaction is a reaction in which a single compound breaks down into two or more simpler substances.

Example:

$$CaCO_3 \xrightarrow{\text{Heat}} CaO + CO_2$$

One reactant — Two products



10. What is redox reaction? Give one example.

Answer:

A redox reaction is a reaction in which oxidation and reduction occur simultaneously.

Example:

$$CuO + H_2 \longrightarrow Cu + H_2O$$

Reduction (CuO loses oxygen) — Oxidation (H_2 gains oxygen)

PART 5 – IMPORTANT QUESTIONS
CHEMICAL REACTIONS AND EQUATIONS

1. What is a chemical reaction?

Answer:

A chemical reaction is a process in which one or more substances (reactants) change into one or more new substances (products) with different chemical properties.

Example:

$$2Mg + O_2 \longrightarrow 2MgO$$



2. Why should a chemical equation be balanced?

Answer:

- To obey the Law of Conservation of Mass.
- Mass can neither be created nor destroyed in a chemical reaction.
- Balanced equations give the exact number of atoms / molecules of each substance.



3. What is the difference between physical change and chemical change? Give one example of each.

Answer:

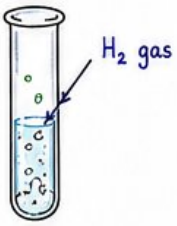
Physical Change	Chemical Change
• No new substance is formed.	• New substance(s) is/are formed.
• Usually reversible.	• Usually irreversible.
• Only physical properties change.	• Chemical properties change.
Example: Melting of ice	Example: Rusting of iron

4. What happens when zinc granules are added to dilute sulphuric acid? Write the balanced equation.

Answer:

- Brisk effervescence occurs.
- A colourless, odourless gas (hydrogen) is evolved.
- The test tube / beaker becomes warm.
- Zinc sulphate is formed.

$$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2\uparrow$$



5. What is a precipitation reaction? Write one example.

Answer:

A reaction in which an insoluble solid (precipitate) is formed is called a precipitation reaction.

Example:

$$Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4\downarrow + 2NaCl$$

(White precipitate of $BaSO_4$ is formed)



6. What are exothermic and endothermic reactions? Give one example of each.

Answer:



Exothermic reaction: A reaction in which heat is released.

Example: $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O + \text{Heat}$



Endothermic reaction: A reaction in which heat is absorbed.

Example: $CaCO_3 \xrightarrow{\text{Heat}} CaO + CO_2$

7. What is oxidation and reduction? Give one example of each.

Answer:

Oxidation: Addition of oxygen or removal of hydrogen.

Example: $2Cu + O_2 \longrightarrow 2CuO$ (Copper is oxidised)

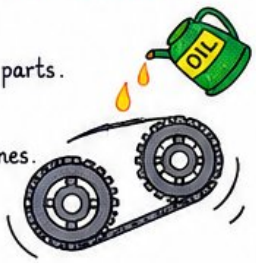
Reduction: Removal of oxygen or addition of hydrogen.

Example: $CuO + H_2 \longrightarrow Cu + H_2O$ (CuO is reduced)

8. Why do we apply oil or grease on machines?

Answer:

- To reduce friction between moving parts.
- To prevent wear and tear.
- To increase the life of the machines.



9. Why does food spoil on exposure to air?

Answer:

- Food contains oil and fats.
- On exposure to air, these get oxidised.
- This causes bad smell, bad taste and discolouration.
- Hence, food gets spoiled (rancidity occurs).



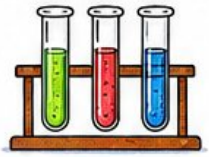
10. Identify the type of reaction.

(a) $2K + 2H_2O \longrightarrow 2KOH + H_2$ (b) $CaCO_3 \longrightarrow CaO + CO_2$

Answer:

(a) This is a Displacement Reaction.

(b) This is a Decomposition Reaction.



PART 6 – MORE IMPORTANT QUESTIONS
CHEMICAL REACTIONS AND EQUATIONS

11. Why is rusting considered a chemical change?

Answer:

- New substance (rust – iron oxide) is formed.
- It is irreversible.
- The chemical properties of iron change.
- Hence, rusting is a chemical change.



$$4Fe + 3O_2 + xH_2O \longrightarrow 2Fe_2O_3 \cdot xH_2O$$
(Rust)

12. Why are food packets flushed with nitrogen gas?

Answer:

- Nitrogen is an inert gas.
- It prevents oxidation of oils and fats present in food.
- It prevents rancidity (spoilage) of food.
- Hence, it increases the shelf life of food.



13. What is a combination reaction? Write one example.

Answer:

A combination reaction is a reaction in which two or more substances combine to form a single product.

Example:

$$CaO + H_2O \longrightarrow Ca(OH)_2$$
Reactant Reactant Product



14. Write one example of decomposition reaction caused by heat.

Answer:

$$CaCO_3 \xrightarrow{Heat} CaO + CO_2$$
Reactant Product Product



15. Give one example of displacement reaction.

Answer:

$$Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$$
More reactive (Zinc) Less reactant Product Displaced copper

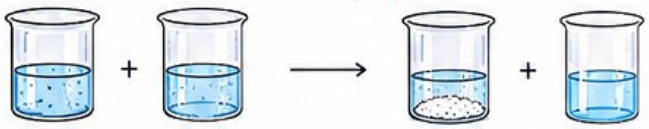


Here, zinc (more reactive metal) displaces copper (less reactive metal) from copper sulphate solution.

16. Give one example of double displacement reaction.

Answer:

$$Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4\downarrow + 2NaCl$$
Reactant Reactant White precipitate Product



17. What is the difference between endothermic and exothermic reactions?

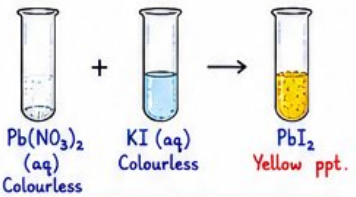
Answer:

Exothermic Reaction	Endothermic Reaction
• Heat is released.	• Heat is absorbed.
• Surroundings become warm.	• Surroundings become cool.
• Example: $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O + Heat$	• Example: $CaCO_3 \xrightarrow{Heat} CaO + CO_2$

18. What are the observations when lead nitrate solution is mixed with potassium iodide solution? Write the chemical equation.

Answer:

- A bright yellow precipitate of lead iodide is formed.
- The solution becomes colourless.



$$Pb(NO_3)_2\,(aq) + 2KI\,(aq) \longrightarrow PbI_2\,(s)\downarrow + 2KNO_3\,(aq)$$

19. Why do we use balance in chemical equations?

Answer:

- To obey the Law of Conservation of Mass.
- Mass can neither be created nor destroyed.
- It ensures that the number of atoms of each element is same on both sides.
- It gives correct stoichiometry (quantitative information) of reactants and products.

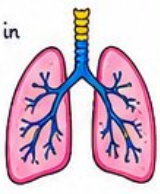


20. What is respiration? Write its balanced equation.

Answer:

Respiration is the process in which glucose is oxidised in the presence of oxygen in our body to release energy.

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + Energy$$
Glucose Oxygen Carbon dioxide Water Energy released



Learn these questions well – they are very important for exams!

PART 7 – MORE IMPORTANT QUESTIONS
CHEMICAL REACTIONS AND EQUATIONS

21. What is oxidation? Give one example.

Answer:

Oxidation is the addition of oxygen or removal of hydrogen.

Example:

$$2Cu + O_2 \longrightarrow 2CuO$$

(Copper is oxidised to copper(II) oxide)



22. What is reduction? Give one example.

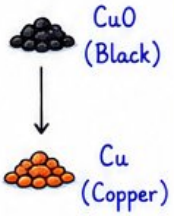
Answer:

Reduction is the removal of oxygen or addition of hydrogen.

Example:

$$CuO + H_2 \longrightarrow Cu + H_2O$$

(Copper(II) oxide is reduced to copper)



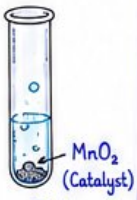
23. What is a redox reaction? Give one example.

Answer:

A redox reaction is a reaction in which both oxidation and reduction occur simultaneously.

Example:

$$\underset{\text{Reduced (CuO loses O)}}{CuO} + \underset{\text{Oxidised (H_2 gains O)}}{H_2} \longrightarrow Cu + H_2O$$



24. State the law of conservation of mass.

Answer:

The law of conservation of mass states that mass can neither be created nor destroyed in a chemical reaction.

Total mass of reactants = Total mass of products



25. What is the role of a catalyst in a chemical reaction?

Answer:

- A catalyst increases the speed of a chemical reaction.
- It does not get consumed in the reaction and remains unchanged.

Example:

MnO_2 is used as a catalyst in the decomposition of hydrogen peroxide.



26. What is the difference between a reversible reaction and an irreversible reaction?

Reversible Reaction	Irreversible Reaction
• The reaction can proceed in both forward and backward directions.	• The reaction proceeds only in the forward direction.
• Products can react again to form reactants.	• Products do not react again to form reactants.
Example: $N_2 + 3H_2 \rightleftarrows 2NH_3$	• Example: $HCl + NaOH \rightarrow NaCl + H_2O$

27. Why do we generally use dilute acids in chemical reactions?

Answer:

- Dilute acids are safer to handle.
- They react slowly and in a controlled manner.
- They produce desired products without too much heat.

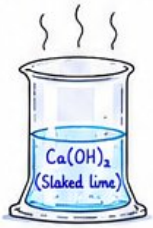


28. What happens when calcium oxide is added to water? Write the chemical equation.

Answer:

- Calcium oxide reacts with water vigorously.
- A large amount of heat is released.
- Calcium hydroxide (slaked lime) is formed.
- Chemical equation:

$$CaO + H_2O \longrightarrow Ca(OH)_2 + \text{Heat}$$



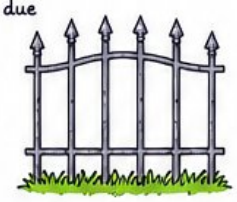
29. What is corrosion? How can it be prevented?

Answer:

Corrosion is the slow destruction of metals due to chemical reaction with air, moisture or other substances.

Prevention:

- Painting
- Oiling / Greasing
- Galvanisation
- Alloying



30. Why are thermite reactions used in welding railway tracks?

Answer:

- Thermite reactions produce a large amount of heat.
- This heat is enough to melt iron.
- The molten iron joins the two ends of the railway tracks.



31. Write balanced chemical equations for the following word equations.

(a) Iron + Sulphur ⟶ Iron sulphide
$Fe + S \rightarrow FeS$

(b) Magnesium + Oxygen ⟶ Magnesium oxide
$2Mg + O_2 \rightarrow 2MgO$

(c) Carbon + Oxygen ⟶ Carbon dioxide
$C + O_2 \rightarrow CO_2$

(d) Aluminium + Copper(II) oxide ⟶ Aluminium oxide + Copper
$2Al + 3CuO \rightarrow Al_2O_3 + 3Cu$

(e) Sodium + Water ⟶ Sodium hydroxide + Hydrogen
$2Na + 2H_2O \rightarrow 2NaOH + H_2$

(f) Barium chloride + Sodium sulphate ⟶ Barium sulphate + Sodium chloride
$BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$

PART 8 – MORE IMPORTANT QUESTIONS
CHEMICAL REACTIONS AND EQUATIONS

32. What is the difference between combination reaction and decomposition reaction?

Answer:

Combination Reaction	Decomposition Reaction
• Two or more substances combine to form a single product. • It is opposite to decomposition reaction.	• A single compound breaks down into two or more simpler substances. • It is opposite to combination reaction.
Example: $CaO + H_2O \rightarrow Ca(OH)_2$ (Reactant, Reactant, Product)	Example: $CaCO_3 \xrightarrow{Heat} CaO + CO_2$ (Reactant, Product, Product)

33. What is the role of oxygen in burning of magnesium?

Answer:

- Magnesium reacts with oxygen present in air.
- It produces magnesium oxide.
- A bright white light and heat are produced.



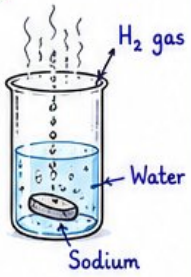
Balanced equation:

$$2Mg + O_2 \rightarrow 2MgO$$

34. What happens when sodium is added to water? Give the balanced chemical equation.

Answer:

- Sodium reacts vigorously with water.
- It produces sodium hydroxide and hydrogen gas.
- Heat is released.



Balanced equation:

$$2Na + 2H_2O \rightarrow 2NaOH + H_2\uparrow$$

35. What is the difference between physical mixture and chemical reaction?

Answer:

Physical Mixture	Chemical Reaction
• No new substance is formed. • The components retain their properties. • Usually, it is reversible. • Example: Sand + Salt	• New substance(s) is/are formed. • The properties of substances change. • Usually, it is irreversible. • Example: Rusting of iron

36. Why is rusting of iron considered a slow chemical reaction?

Answer:

- Iron reacts very slowly with oxygen and moisture present in the air.
- The reaction takes place over a long time.
- New substance (rust) is formed.

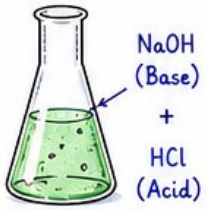


$$4Fe + 3O_2 + xH_2O \rightarrow 2Fe_2O_3 \cdot xH_2O \text{ (Rust)}$$

37. Write balanced equation for the neutralisation reaction between sodium hydroxide and hydrochloric acid.

Answer:

- Sodium hydroxide (base) reacts with hydrochloric acid to form sodium chloride and water.
- This reaction is called neutralisation reaction.



Balanced equation:

$$NaOH + HCl \rightarrow NaCl + H_2O$$

38. Why is photosynthesis considered an endothermic process?

Answer:

- During photosynthesis, plants absorb sunlight energy.
- This energy is used to convert carbon dioxide and water into glucose.
- Since energy is absorbed, it is an endothermic process.



$$6CO_2 + 6H_2O \xrightarrow{Sunlight} C_6H_{12}O_6 + 6O_2$$

39. What are the effects of corrosion?

Answer:

- It weakens metals.
- It causes damage to bridges, buildings, machines, vehicles, etc.
- It leads to economic loss.
- It reduces the life of metal articles.



40. Classify the following reactions into combination, decomposition, displacement and double displacement reactions. Write balanced chemical equations.

(a) Calcium carbonate ⟶ Calcium oxide + Carbon dioxide

Type: Decomposition reaction

Balanced equation:

$$CaCO_3 \xrightarrow{Heat} CaO + CO_2$$



(b) Zinc + Copper(II) sulphate ⟶ Zinc sulphate + Copper

Type: Displacement reaction

Balanced equation:

$$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$$



(c) Barium chloride + Sodium sulphate ⟶ Barium sulphate + Sodium chloride

Type: Double displacement reaction

Balanced equation:

$$BaCl_2 + Na_2SO_4 \rightarrow BaSO_4\downarrow + 2NaCl$$

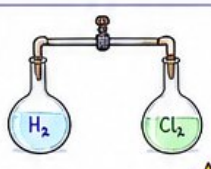


(d) Hydrogen + Chlorine ⟶ Hydrogen chloride

Type: Combination reaction

Balanced equation:

$$H_2 + Cl_2 \rightarrow 2HCl$$



Practice these questions regularly. They are very important for exams!

PART 9 – MORE IMPORTANT QUESTIONS
CHEMICAL REACTIONS AND EQUATIONS

41. What is bleaching powder? Write its chemical formula and one use.

Answer:

- Bleaching powder is a compound used for bleaching clothes and for disinfecting water.
- Chemical formula: $CaOCl_2$
- One use: It is used to disinfect drinking water.



42. Why is calcium oxide called a basic oxide? Write the chemical equation for its reaction with water.

Answer:

- Calcium oxide is called a basic oxide because it reacts with water to form a base (calcium hydroxide).
- Chemical equation:

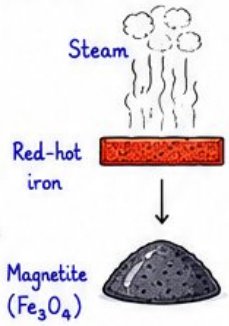
$$\underset{\text{(Basic oxide)}}{CaO} + H_2O \longrightarrow \underset{\text{(Base)}}{Ca(OH)_2}$$



43. What is the effect of adding steam to red-hot iron?

Answer:

- When steam is passed over red-hot iron, a layer of magnetite (Fe_3O_4) is formed.
- This process is used in the manufacture of magnetite.



44. Write balanced chemical equation for the reaction between barium hydroxide and aluminium sulphate.

Answer:

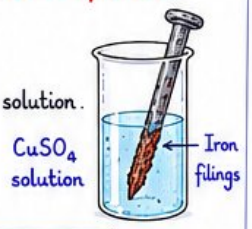
$$\underset{\text{Barium hydroxide}}{3Ba(OH)_2} + \underset{\text{Aluminium sulphate}}{Al_2(SO_4)_3} \longrightarrow \underset{\substack{\text{Barium sulphate}\\\text{(White ppt.)}}}{3BaSO_4\downarrow} + \underset{\substack{\text{Aluminium hydroxide}\\\text{(White ppt.)}}}{2Al(OH)_3\downarrow}$$



45. What happens when iron filings are added to copper sulphate solution? Write the balanced equation.

Answer:

- Iron is more reactive than copper.
- It displaces copper from copper sulphate solution.
- Reddish-brown copper is deposited on iron filings and green colour of the solution fades.



$$\underset{\text{(Iron)}}{Fe} + \underset{\text{(Copper sulphate)}}{CuSO_4} \longrightarrow \underset{\text{(Iron sulphate)}}{FeSO_4} + \underset{\text{(Copper)}}{Cu}$$

46. Write balanced chemical equation for the combustion of methane.

Answer:

- Methane burns in oxygen to form carbon dioxide and water with release of heat.

Balanced equation:

$$\underset{\text{Methane}}{CH_4} + \underset{\text{Oxygen}}{2O_2} \longrightarrow \underset{\text{Carbon dioxide}}{CO_2} + \underset{\text{Water}}{2H_2O} + \underset{\text{Heat released}}{\text{Heat}}$$

47. What is the difference between rusting and corrosion?

Answer:

Rusting	Corrosion
• It occurs only in iron. • It forms a reddish-brown flaky substance ($Fe_2O_3 \cdot xH_2O$). • It is a specific type of corrosion.	• It occurs in all metals. • It can form different products depending on the metal. • Rusting is one example of corrosion.

48. What is the role of carbon dioxide in a fire extinguisher?

Answer:

- CO_2 does not support burning.
- It is heavier than air and forms a blanket over the fire.
- It cuts off the supply of oxygen and helps in extinguishing the fire.



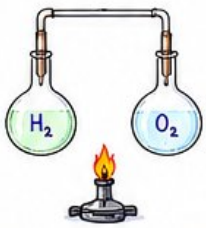
49. Write balanced chemical equation for the reaction between hydrogen and oxygen.

Answer:

- Hydrogen burns in oxygen to form water.

Balanced equation:

$$\underset{\text{Hydrogen}}{2H_2} + \underset{\text{Oxygen}}{O_2} \longrightarrow \underset{\text{Water}}{2H_2O}$$



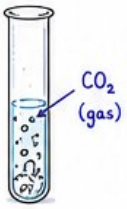
50. What are the observations when dilute acetic acid is added to sodium hydrogen carbonate? Write the balanced equation.

Answer:

- Brisk effervescence (bubbles) of CO_2 is produced.
- A colourless gas is evolved.
- The test tube becomes cool.

Balanced equation:

$$CH_3COOH + NaHCO_3 \longrightarrow CH_3COONa + H_2O + CO_2\uparrow$$



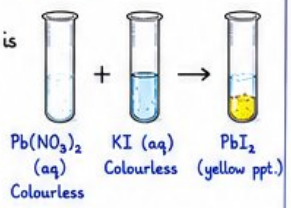
These questions are very important for your exams. Revise them well and practice regularly!

PART 10 – MORE IMPORTANT QUESTIONS
CHEMICAL REACTIONS AND EQUATIONS

51. What happens when lead nitrate solution is mixed with potassium iodide solution? Write the balanced chemical equation.

Answer:

- A bright yellow precipitate of lead iodide is formed.
- The solution becomes colourless.
- Lead iodide is insoluble in water.



Balanced equation:

$$Pb(NO_3)_2\ (aq) + 2KI\ (aq) \longrightarrow PbI_2\ (s)\downarrow + 2KNO_3\ (aq)$$

(Yellow ppt.)

52. Why do we store food in refrigerator? Give the reason.

Answer:

- Low temperature in a refrigerator slows down the chemical reactions.
- It prevents spoilage and rancidity of food.
- It increases the shelf life of food.



53. What is rancidity? How can it be prevented?

Answer:

Rancidity is the process of oxidation of oils and fats present in food, causing bad taste and smell.

Prevention:

- By adding antioxidants (like BHA).
- By flushing food packets with nitrogen gas.
- By storing food in airtight containers.



54. What causes corrosion of metals? How can it be prevented?

Answer:

Corrosion is caused by the reaction of metals with substances like air, moisture or chemicals.

Prevention:

- Painting
- Oiling / Greasing
- Galvanisation
- Electroplating
- Alloying



55. What is the use of baking soda?

Answer:

- Baking soda (sodium hydrogen carbonate) is used in making cakes and bread.
- It releases CO_2 gas when heated.
- The CO_2 gas makes the cake or bread soft and spongy.
- It is also used as an antacid.



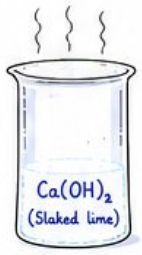
Chemical equation (when heated):

$$2NaHCO_3 \xrightarrow{Heat} Na_2CO_3 + CO_2\uparrow + H_2O$$

56. Why is slaking of lime an exothermic reaction? Write the chemical equation.

Answer:

- When quicklime (CaO) reacts with water, a large amount of heat is released.
- Hence, slaking of lime is an exothermic reaction.



Chemical equation:

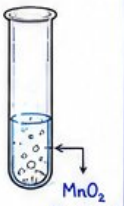
$$CaO + H_2O \longrightarrow Ca(OH)_2 + Heat$$

57. What is the role of a catalyst in a chemical reaction? Give one example.

Answer:

- A catalyst increases the speed of a chemical reaction.
- It does not get used up in the reaction.
- It remains chemically unchanged.

Example: MnO_2 is used as a catalyst in the decomposition of hydrogen peroxide.



Chemical equation:

$$2H_2O_2 \xrightarrow{MnO_2} 2H_2O + O_2\uparrow$$

58. What is the difference between decomposition reaction and double displacement reaction? Give one example of each.

Decomposition Reaction	Double Displacement Reaction
• A single compound breaks down into two or more simpler substances. • Example: $CaCO_3 \xrightarrow{Heat} CaO + CO_2$	• Two compounds exchange their ions to form two new compounds. • Example: $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4\downarrow + 2NaCl$

59. Why does a white coating appear on old silver articles kept in the open? How can it be cleaned?

Answer:

- Silver reacts with hydrogen sulphide gas present in air to form silver sulphide (black coating).
- It is then converted to silver carbonate (white coating).
- The white coating can be removed by rubbing with toothpaste or a paste of baking soda.



60. Why do we prefer soap over synthetic detergents for washing clothes?

Answer:

- Soap is biodegradable and does not cause pollution.
- It is gentler on skin.
- Synthetic detergents are non-biodegradable and may cause water pollution.



61. Identify the type of reaction in the following and write one benefit:

(a) $2Fe + 3Cl_2 \longrightarrow 2FeCl_3$

Type: Combination Reaction

Benefit: Used in the manufacture of iron(III) chloride.



(b) $2AgCl \xrightarrow{Sunlight} 2Ag + Cl_2$

Type: Decomposition Reaction

Benefit: Used in photography.



62. Write balanced chemical equations for the following word equations:

(a) Quicklime + Water → Slaked lime

$$CaO + H_2O \longrightarrow Ca(OH)_2$$

(b) Aluminium + Oxygen → Aluminium oxide

$$4Al + 3O_2 \longrightarrow 2Al_2O_3$$

Practice these questions regularly and score full marks in your exams!